



GAME-SPECIFIC FACULTY GUIDE

CPI Live

A consumer-price measurement game built around a fixed basket, CPI, and inflation.

CPI Live asks students to construct a price measure from a household basket. Use their calculations and interpretations to connect the cost of that basket to an index and then to a rate of change.

Why This Game Exists

I wanted students to see what they were constructing when they calculated CPI. It's possible to follow the steps and still be unsure why quantities stay fixed, why the base year is 100, or why subtracting two index values usually isn't the inflation rate.

CPI Live starts with the basket and its prices. Students calculate the cost, build the index, then measure inflation and explain what it means. I wanted each calculation to grow out of the previous one. The weighting comparison, analyst's mistake, and later period give them another chance to use that reasoning.

What Students Do

Students complete ten price checks using a supplied basket of rent, groceries, gas, streaming, and haircuts. The game selects from authored baskets and price changes, so values can differ across runs. Students keep their assigned quantities throughout the calculation; they do not choose a new shopping basket each year. The activity is untimed.

In Year 1, the base year, students calculate the groceries expenditure and total basket cost; the panel supplies the other four expenditures. They then reprice the same quantities at Year 2 prices, calculate CPI, and calculate inflation from Year 1 to Year 2. Verified costs and measures appear in the basket panel, alongside each item's dollar contribution to the change.

Students next interpret a separate index level and predict which of two isolated price changes would move CPI more. After a prediction, the game reveals the dollar effects for comparison. The audit then presents an analyst's draft containing one error. Students identify its source, and correct identification reveals the repaired calculation. They diagnose the calculation rather than type a replacement report.

The final checks use a separate four-year CPI example that leaves their household basket unchanged. Students calculate Year 3 annual inflation, compare annual rates, and interpret a Year 4 decline relative to both the previous year and the base. A CPI line chart, inflation bars, and table distinguish levels from rates; rate information is revealed after the relevant calculation or interpretation attempt.

Students can correct unsuccessful answers and use Need help? for a conceptual hint, with the formula one step deeper. Results summarize the basket, CPI, inflation, largest dollar contribution, and performance across the checks, followed by an interpretation recap. Play Again offers different examples. The separate index-level question asks what CPI establishes relative to the base year, without assuming it gives annual inflation.

What the Game Is Really Teaching

Students keep the same quantities while prices change, so they can compare the cost of the same purchases. Changing quantities too would mix a price change with a change in what was bought.

For each item in the basket, expenditure equals price $\times$ fixed quantity. Adding those expenditures gives the basket cost in dollars. Both price increases and decreases enter the total. Adding unit prices alone would miss how much of each item the household buys.

Students then calculate $CPI = (\text{current basket cost} \div \text{base basket cost}) \times 100$. In the base year, they divide the cost by itself, so CPI is 100 regardless of the basket's dollar cost. Later values compare the cost with that same reference.

Once they have CPI, students use it to measure inflation: $\text{Inflation} = ((\text{new CPI} - \text{previous CPI}) \div \text{previous CPI}) \times 100$. CPI is a level; inflation is its percentage change over an interval. Subtraction alone gives an index-point change. The first annual calculation starts from CPI 100, but the later example uses a different previous-year value. Ask students to name the comparison period before calculating.

Next, students predict which of two isolated price changes would move CPI more, with other prices held at their base levels. An item's effect depends on its base expenditure—quantity times base price—and its percentage price change. A large percentage increase in a small expense can add fewer dollars than a smaller increase in a major expense. Students predict first, then use the revealed contributions to explain the result.

The analyst's draft gives students a method to check. The error can involve changed quantities, equal averaging of item price changes, a reversed CPI ratio, an index level mistaken for inflation, or the wrong annual comparison period. Students must inspect the analyst's method to decide which error occurred. A plausible-looking final number cannot establish that the calculation measures the intended thing.

In the separate timeline, students distinguish rising prices with slower inflation, faster inflation, an unchanged price level, and falling prices. Disinflation means prices still rise more slowly; deflation means the index falls. Students compare the final year's level with both the preceding year and the base year, showing why a decline need not undo the cumulative increase. The two charts keep the CPI level and annual percentage change separate.

CPI Measurement Challenges

After completing the ten core price checks, students can choose *Behind the Index: Where CPI Measurement Gets Complicated*. This optional application has three short interactive episodes. It leaves their original basket, CPI, inflation, score, and completion intact. Students can instead replay the core game or return to Games.

The extension separates the clean fixed-basket teaching model from difficulties encountered when products and consumer choices change. It does not replace the main CPI formula, compute a “true CPI,” or treat current consumer spending as the price index.

Substitution bias

Students see two substitute products, two purchases of each in the base basket, and a relative price increase. They move one purchase toward the cheaper alternative; at least one purchase of the first product must remain. The household is explicitly willing to make that limited substitution while obtaining the meals, fruit portions, or cereal servings it needs. This assumption matters: cheaper spending alone cannot prove unchanged consumer well-being.

For the beef/chicken example, base prices of \$6 and \$4 give a \$20 original basket. New prices of \$10 and \$4 make that same basket cost \$28, an increase of 40%. Moving one beef purchase to chicken gives a \$22 substituted bundle, 10% above the original base cost. Students compare those expenditures and identify which comparison is larger. The fruit variant compares \$8 → \$12 with a \$10 alternative; cereal brands compare \$16 → \$24 with a \$20 alternative. Both give 50% versus 25% growth relative to the original base expenditure.

The feedback names substitution bias after students make the comparison. When consumers can switch toward relatively cheaper substitutes while maintaining suitable consumption, holding the original quantities fixed can overstate cost-of-living growth. The alternative is labeled the cost of a substituted bundle, not another official CPI.

New goods

A visible empty position in the original basket contrasts with a newly available product. Students first decide whether its current price can simply be inserted into the original base-year calculation. It cannot: the product did not exist then and has no base-period price or assigned original quantity. The game never fabricates those data.

Correctly identifying that limitation reveals a purchasing opportunity. Authored examples compare a \$20 streaming plan with a new \$12 service, a \$40 commuter pass with a new \$30 shuttle pass, or a \$30 prescription with a new \$20 approved generic. Each states why the option is suitable for that household. These differences describe potential spending savings, not an estimated index change or an exact welfare gain.

Students then choose a later basket update with an appropriate linked comparison. Feedback explains new goods bias as delayed incorporation and missed consumer gains. It explicitly rejects the idea that CPI excludes new products forever.

Quality change

Students compare an old durable good with an improved replacement. An explicit analyst estimate assigns a dollar value to added measurable quality. Students enter the comparable price, the unadjusted percentage price change, and the quality-adjusted percentage change, using the old price as the denominator for both rates.

Product	Old price	New price	Estimated added quality	Comparable price
Laptop	\$1,000	\$1,100	\$150	\$950
Phone	\$500	\$550	\$75	\$475
Washing machine	\$800	\$880	\$120	\$760

Every authored case gives a raw price increase of 10% and an adjusted comparable-price decline of 5%. For example, $(\$950 - \$1,000) \div \$1,000 \times 100 = -5\%$. Students subtract the stated quality value from the replacement price; they do not divide by a performance score or estimate a regression. These are simplified, internally consistent analyst estimates, not claims that all improvements can be measured exactly.

CPI attempts to isolate price change from quality change. Statistical agencies adjust for measurable differences, but those adjustments require evidence and judgment and are not perfect.

Closing comparison and replay

Three comparison panels show original versus shifted quantities, the old basket versus an expanded choice set, and raw versus quality-adjusted prices. The completed core readouts and the normal CPI and inflation formulas remain available as the foundation. The close emphasizes that CPI is useful: basket updates and quality adjustments reduce measurement problems. The episodes illustrate possible overstatement, not a claim that every CPI error has the same direction or that CPI is inherently wrong.

Replaying the extension offers different contexts for all three measurement problems. Students can compare how the same reasoning applies to another pair of substitutes, new product, or quality change. Extension activity leaves the ten-check core score unchanged.

For discussion, ask which assumption makes the substitution comparison meaningful, what base-period information is missing for a new good, and why a higher replacement price may not all be inflation. Require students to distinguish each illustrative comparison from an official index estimate.

For faculty background, see the Bureau of Labor Statistics on [assessing and improving CPI accuracy](#) and [quality adjustment](#). These explain why official measurement uses methods beyond the game's simplified fixed-basket model.

The Educational Loop

Students price the basket, observe its changing cost, and use feedback to explain or correct their calculations. CPI and inflation formalize those measurements. The weighting prediction, analyst audit, and separate multi-year example ask students to apply the method and interpret what its numbers can establish. The final recap brings the distinctions together.

Experience → Consequence → Economic Explanation → Formalization → Transfer

Using It in Class

These are possible starting points; choose what fits your course.

Before formal CPI instruction

Let students calculate the two basket costs first. Pause before the index calculation and derive the CPI formula from the comparison they have already made.

During CPI instruction

Use a run after introducing the fixed-basket idea. Before students calculate CPI or inflation, ask what each denominator represents and which periods they are comparing.

After instruction

Ask students to explain an audit diagnosis or connect a change in the CPI line to the corresponding inflation bar. Use the weighting comparison to check whether they consider expenditure shares.

Independent/student review

Students can use hints, feedback, and retries through the final interpretation. You can request a reflection or results screenshot through your usual course workflow; the game sends no automatic faculty report.

Questions to Consider

Select any of these prompts for discussion or a short reflection.

1. What would become ambiguous if you changed the basket quantities while comparing Year 1 and Year 2 prices?
2. Why does every assigned basket have a base CPI of 100 even when the basket costs differ?
3. What distinguishes the basket's dollar cost from its CPI? What does each number let you compare?
4. Why can subtracting 100 appear to work in the first inflation calculation but fail as a method for later annual comparisons?
5. In the weighting comparison, how did base expenditure and percentage price change combine to determine the larger effect?
6. Which part of the analyst's method identified the error? How would you explain why the repaired method answers the intended question?
7. How can the CPI line rise while annual inflation falls? What would the line and bars show if prices actually fell?
8. When a later CPI declines, what comparisons tell you whether prices have returned to the base-year level? Why is the latest annual rate insufficient on its own?

Common Misconceptions

- “CPI is the basket's dollar price.” Compare the dollar total with the ratio that expresses it relative to the base year.
- “The base CPI depends on how expensive the basket is.” The base cost is divided by itself, giving an index of 100.
- “CPI tells me this year's inflation by itself.” Identify the previous year's index and calculate the percentage change over that interval.
- “Subtracting two index values gives the inflation rate.” Divide the difference by the previous CPI and multiply by 100.
- “The largest percentage price increase matters most.” Check each item's base expenditure and the dollars its price change adds to the basket.
- “Lower inflation means prices fell.” Distinguish a slower positive rate from a negative rate, and compare the CPI levels.